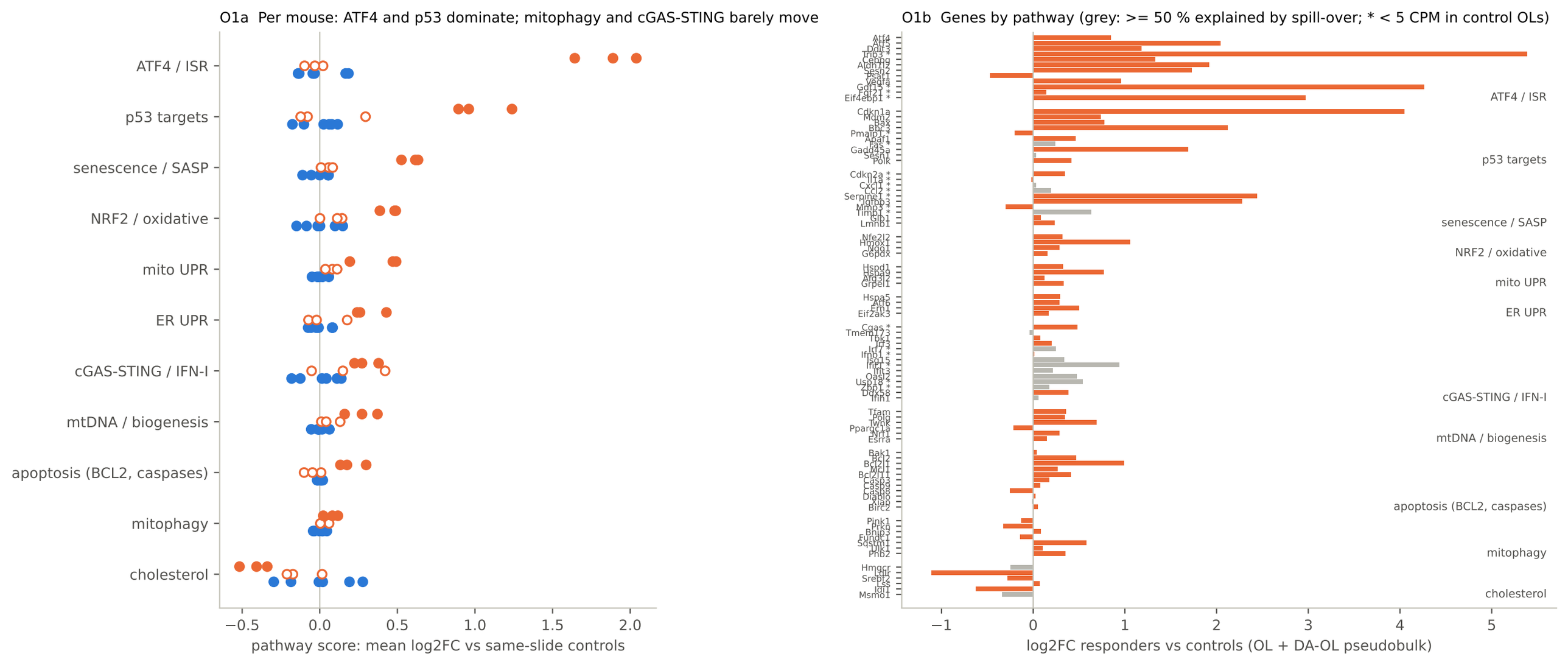
O1a Per mouse: ATF4 and p53 dominate; mitophagy and cGAS-STING barely move
ATF4 / ISR
p53 targets
senescence / SASP
NRF2 / oxidative
mito UPR
ER UPR
cGAS-STING / IFN-I
mtDNA / biogenesis
apoptosis (BCL2, caspases)
mitophagy
cholesterol
−0.5
0.0
0.5
1.0
1.5
2.0
pathway score: mean log2FC vs same-slide controls
O1b Genes by pathway (grey: >= 50 % explained by spill-over; * < 5 CPM in control OLs)
Atf4
Atf3
Ddit3 *
Cebpg
Atgrip1
Sesn2
Psat1
Vegfa
Gdf15 *
Ppp1r15a *
Eif4ebp1 *
Cdkn1a
Mdm2
Bax
Bbc3
Pmaip1 *
Aen
Gadd45a
Sesn1
Polk
Cdkn2a *
Il1a *
Cxcl1 *
Ccl2 *
Serpine1 *
Ldhb
Mmp3 *
Timp1 *
Gbp1
Lmnb1
Nfe2l2
Hmox1
Nqo1
Gpx4
Hspd1
Hspa9
Atg3
Grpel1
Hspa5
Atf6
Ern1
Eif2ak3
Cgas *
Tmem173
Tbk1
Irf3
Irf7 *
Ifnb1 *
Isg15
Ifit1 *
Ifit3
Oasl2
Usp18 *
Zbp1 *
Ddx58
Ifih1
Tfam
Polg
Twnk
Ppargc1a
Nrf1
Esrra
Bak1
Bcl2
Bcl2l1
Mcl1
Bcl2l11
Casp3
Casp9
Casp8
Diablo
Xiap
Birc2
Pink1
Prkn
Bnip3
Fundc1
Sqstm1
Tbk1
Pnb2
Hmgcr
Idi1
Srebf2
Fdps
Idi1
Msmo1
ATF4 / ISR
p53 targets
senescence / SASP
NRF2 / oxidative
mito UPR
ER UPR
cGAS-STING / IFN-I
mtDNA / biogenesis
apoptosis (BCL2, caspases)
mitophagy
cholesterol
−1
0
1
2
3
4
5
log2FC responders vs controls (OL + DA-OL pseudobulk)